

BWS44406: Industry Integrated Project Framework and Development

Synopsis:

This course will expose the students on the process of conducting research in order to provide the skills and ability in carrying out research project framework and development in the data analytical field. The covered areas consisting of research design, project management, design of innovation, data collection and analysis. All the works can consider to publish via conference and journal publication.

Course Learning Outcomes (CLO):

At the end of this course, the students will be able to:

CLO1: Describe a solution to data analytical problems using a systematic research methodology effectively. (C5; LOD3; PLO2)

CLO2: Use appropriate and relevant resources to develop a project based on organization needs. (P5; LOD13; PLO9)

CLO3: Practise good ethics in managing the project. (A4; LOD18; PLO11)

Content Outline of the Course:

TOPIC 1: Research Design

- i. Determine the type of research/convention selected to conduct the study.
- ii. Sequence the different stages and link them with data collection and outputs.
- iii. Evaluate the strengths and weaknesses of construction methods.

TOPIC 2: Construction Innovation

- i. Learn how to handle an integrated project based on the chosen project type.
- ii. Explore types of innovations related to current industry practices.
- iii. Assess the value of creative solutions.
- iv. Understand basic innovation management principles.

TOPIC 3: Project Delivery Methods

- i. Understand the fundamentals of project delivery methods.
- ii. Apply design-build methods to implement projects.
- iii. Select appropriate delivery methods for different project types.

TOPIC 4: Data Collection and Analysis

- i. Collect and manage primary and secondary quantitative data sets.
- ii. Analyze data, interpret findings, and validate results.

TOPIC 4: Attributes of Framework

- i. Understand requirements for conferences related to the project.
- ii. Prepare materials suitable for publication and professional dissemination

Method of assessment: Final Year Project **Materials to deliver the course:** Lecture Slide, Gantt charts and project scheduling tools, project work, practical session, case studies, meeting with university supervisor, etc.

Expected outcomes:

Students should be able to develop and execute project or research approaches, design effective research projects, collect and analyze data proficiently, and understand and apply the attributes of relevant frameworks to their work.

Assessment method: Interview / Progress Report/ Presentation / Demonstration / Practical Task/ Reflective Log

Assessment

PI Q2: Cognitive Skill **PI Q9:** Personal Skill **PI Q11:** Ethics and Professionalism

Assessment	Assessor	CLO	PLO	Level	Tool	Assessment	%
Continuous	University	CLO1	PLO2	C5	WBL Logbook Progress Report		20
		CLO3	PLO11	A5	Video Presentation		25
	Industry	CLO1	PLO2	C5	WBL Logbook Progress Report		15
		CLO2	PLO9	P5	Observation		40
Total							100